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Quanta in the Sky with Diamonds

INFLATION, QUANTUM JITTERS, AND THE ARROW OF TIME

The discovery of the inflationary framework launched a new era in cosmological research, and in the decades since, many thousands of papers have been written on the subject. Scientists have explored just about every nook and cranny of the theory you could possibly imagine. While many of these works have focused on details of technical importance, others have gone further and shown how inflation not only solves specific cosmological problems beyond the reach of the standard big bang, but also provides powerful new approaches to a number of age-old questions. Of these, there are three developments—having to do with the formation of clumpy structures such as galaxies; the amount of energy required to spawn the universe we see; and (of prime importance to our story) the origin of time's arrow—on which inflation has ushered in substantial and, some would say, spectacular progress.

Let's take a look.

Quantum Skywriting

Inflationary cosmology's solution to the horizon and flatness problems was its initial claim to fame, and rightly so. As we've seen, these were major accomplishments. But in the years since, many physicists have come to believe that another of inflation's achievements shares the top spot on the list of the theory's most important contributions.

The lauded insight concerns an issue that, to this point, I have encouraged you not to think about: How is it that there are galaxies, stars, planets, and other clumpy things in the universe? In the last three chapters, I asked you to focus on astronomically large scales—scales on which the universe appears homogeneous, scales so large that entire